

# Hein Zarni Aung

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## EDUCATION

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**Stony Brook University** — Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Software Security, Systems Fundamentals I & II, Algorithms

## RESEARCH

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**Undergraduate Research Assistant — Ethos Lab, SBU** — Sep 2025 – Dec 2025 (paused)

PI: Abisheka Pitumpe | Team size: 2

- Built a real-time monitoring pipeline for newly issued SSL/TLS certificates using Selenium and Python to scrape, filter, and flag fraudulent domains, designing heuristic-based detection logic with a faculty PI.

## INTERNSHIPS

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**Software Engineering Intern — Clustr** — Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built and owned server-side integrations using Firebase Cloud Functions — CRUD APIs for communities, posts, and comments with enforced permission checks server-side, plus join/leave workflows and activity feed endpoints, shipped end-to-end into production.
- Contributed 50+ commits across backend and frontend features, collaborating with an 8-person, multi-developer team via Git to keep community and scheduling modules in sync with the shared codebase.
- Implemented secure authentication (password reset flow, custom claims for onboarding) and a Scheduler UI module with custom date/time picker and calendar-completion animation.

**Software Engineering Intern — Prenora Dynamics LLC** — Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).

## CAMPUS INVOLVEMENT

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**International Peer Mentor — Office of Global Affairs, Stony Brook University** — Aug 2025 – Present

- Mentored 6 international students through biweekly one-on-one meetings, providing hands-on guidance on academic strategies, communication, and cross-cultural adjustment.

## PROJECTS

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**TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon)** — Apr 2026

Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4

- Built a data pipeline extracting pitch, rhythm, and tone features from audio using Essentia/NumPy to power an AI-matching tool pairing music students with instructors based on technical weaknesses.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Collaborated in a 4-person team; broader app included a React/Tailwind frontend, Python/FastAPI backend in Docker, Cloud Firestore, GCP hosting, and ElevenLabs-generated voice feedback reports.

## SKILLS

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**Languages:** Python, TypeScript, JavaScript, Java, C

**Backend:** Node.js, Express.js, FastAPI

**Databases & Cloud:** PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

**Frontend:** React, React Native, Next.js

**Tools:** Git, Docker, Linux, PM2, Selenium